CHURNGUARD AI

Customer Churn Prediction & Retention Intelligence System

Project Report · MainCrafts 2-Day Skill Certification Program

Domain: Artificial Intelligence & Machine Learning

Candidate: Your Name

Date: 30 September 2026

Environment: Python · pandas · scikit-learn · Google Colab / Jupyter

Headline result (unseen test customers)	
Selected model	Logistic Reg. + features + class weight
Recall / Precision / F1	79.3% / 59.5% / 0.680
ROC-AUC	0.877
Dataset	Telco Customer Churn (IBM schema) - schema-identical replica — 7,043 customers

1. Project Title

CHURNGUARD AI — Customer Churn Prediction & Retention Intelligence System. A supervised machine-learning classification workflow that predicts whether a telecom customer is likely to churn.

2. Problem Statement

Businesses lose customers for many reasons, and they usually find out after the customer has already decided to leave. At that point retention is expensive or impossible. The problem addressed here is to learn, from historical customer records, which customers are likely to churn so that the company can act before the customer leaves.

3. Objective

- Analyse customer data and identify the patterns that separate customers who churn from those who stay.
- Prepare the data (cleaning, encoding, scaling, feature engineering) inside leak-free scikit-learn pipelines.
- Train and compare at least two classification models and select one with a justified metric (recall / F1, not accuracy alone).
- Predict churn probability and a HIGH / MEDIUM / LOW risk level for new customers.
- Identify the features that influence predictions and turn the findings into practical business recommendations.

4. Dataset Description

The project uses the Telco Customer Churn schema (IBM sample dataset): 7,043 customers and 21 columns. The target is Churn (Yes/No); the remaining columns describe demographics (gender, senior citizen, partner, dependents), account information (tenure, contract, payment method, paperless billing, monthly and total charges) and subscribed services (phone, internet, security, backup, protection, support, streaming). Overall churn rate: 28.2% (an imbalanced problem).

Important note on data provenance. The official IBM CSV could not be downloaded in the environment where this project was assembled (no internet access). The file in dataset/customer_churn.csv is therefore a schema-identical replica (same 21 columns, value formats and data-quality quirks such as blank TotalCharges for new customers) generated by dataset/generate_synthetic_dataset.py. All figures in this report belong to that file. Replacing the CSV with the official IBM / MainCrafts file and re-running the notebook and report/generate_report.py regenerates every number, chart and statement without any code change.

Property	Value
Records	7,043
Columns (raw)	21
Input features used	19 original (+ 6 engineered)
Numerical features	SeniorCitizen, tenure, MonthlyCharges, TotalCharges
Categorical features	15 columns (contract, internet service, payment method, ...)
Target	Churn (Yes = 1, No = 0)
Missing / blank values	11 blank TotalCharges strings (tenure = 0); 0 NaN before conversion

Property	Value
Duplicate records	0
Train / test split	5,634 / 1,409 (80 / 20, stratified, random_state = 42)

Dataset preview (first 5 rows, selected columns)

customerID	tenure	Contract	InternetService	PaymentMethod	Monthly Charges	TotalCharges	Churn
9774-KWRWK	26	Month-to-month	DSL	Electronic check	45.12	1163.93	Yes
4747-RIFIN	33	Two year	DSL	Electronic check	74.17	2424.78	No
9019-EBASM	14	Month-to-month	Fiber optic	Electronic check	108.0	1554.36	Yes
7340-WDOHS	17	Month-to-month	Fiber optic	Bank transfer (automatic)	92.36	1579.53	Yes
4663-QQAUO	13	Month-to-month	No	Electronic check	18.25	231.04	Yes

5. Technologies Used

Tool	Purpose
Python 3	Programming language
Google Colab / Jupyter	Free hosted notebook environment
pandas, NumPy	Data loading, cleaning, aggregation, numeric operations
Matplotlib, Seaborn	Exploratory analysis and result visualisations
scikit-learn	Train/test split, OneHotEncoder, StandardScaler, ColumnTransformer, Pipeline, models, GridSearchCV, permutation importance, metrics
joblib	Saving / loading the trained model
ReportLab	Generating this PDF report
Streamlit (optional)	Web interface — CHURNGUARD AI Customer Risk Predictor

6. Methodology

The notebook follows the required end-to-end workflow and explains why each step is done:

Stage	What was done	Why
1. Data understanding	Shape, dtypes, missing values, duplicates, statistics, target definition	Know the data before changing it; find hidden problems
2. Cleaning	TotalCharges to numeric; blanks filled with 0; ID dropped; categories inspected	Models need clean numeric input; no leakage of identifiers
3. Exploration	6 visualisations, each with a written conclusion	Discover churn drivers and class imbalance
4. Feature preparation	Stratified 80/20 split; one-hot encoding in a pipeline; engineered features	Test set must represent unseen data; pipelines prevent leakage
5. Model training	Logistic Regression and Random Forest baselines, then improved variants	Compare a simple and a complex model
6. Evaluation	Accuracy, precision, recall, F1, ROC-AUC, confusion matrix, 5-fold CV	Accuracy is misleading on imbalanced churn data
7. Prediction	predict_churn() returns probability, risk level and suggested action	Make the model usable for new customers
8. Business insight	Risk bands, feature importance, recommendations, limitations	Turn predictions into decisions

7. Data Preprocessing

Issue found	Action	Justification
TotalCharges stored as text (blank strings)	pd.to_numeric(errors='coerce')	Column is numeric by meaning
11 missing TotalCharges	Filled with 0 (not deleted, not mean)	All these customers have tenure = 0, i.e. nothing billed yet; deleting rows loses real customers, the mean would invent spend
customerID	Dropped	Unique identifier, no predictive meaning
Duplicate rows (0)	Checked before dropping the ID; drop_duplicates()	Avoid double-counting
'No internet service' / 'No phone service'	Kept as separate categories	They carry real information (no such service)
Categorical text columns	OneHotEncoder(handle_unknown= 'ignore') in a ColumnTransformer	Models need numbers; unknown categories will not crash prediction
Numeric scale differences	StandardScaler (Day 2 pipelines)	Removes the optimiser ConvergenceWarning of the unscaled baseline
Class imbalance	class_weight ('balanced')	Missing a churner is the costly error

Engineered features: num_addons, protection_services, avg_monthly_spend, charge_vs_avg, tenure_group, has_family.

8. Exploratory Analysis

Key observations drawn from the actual data (each is also written under its chart in the notebook):

- Contract type is the strongest business lever: month-to-month customers churn at 45.9% versus 3.5% on two-year contracts.
- Churn is concentrated among new customers: 51.4% churn in the first 12 months versus 4.4% after 4 years.
- Churners have a median tenure of 15 months versus 35 months for customers who stayed.
- Churners pay more per month on average (₹77.01 vs ₹58.80).
- By internet service, churn is highest for Fiber optic (42.2%); by payment method it is highest for Electronic check (32.6%).

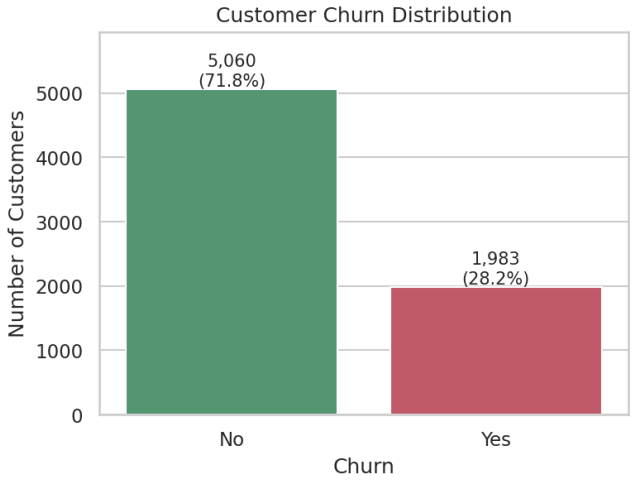


Figure 1 — Churn distribution: the classes are imbalanced, so accuracy alone is misleading.

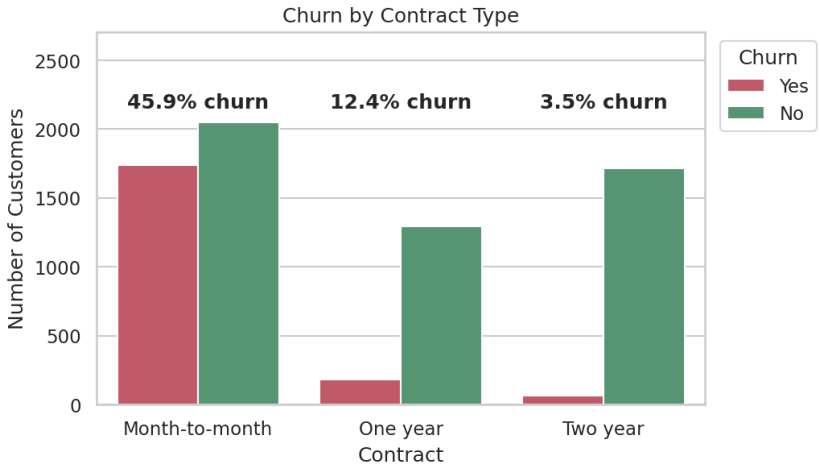


Figure 2 — Churn vs contract type.

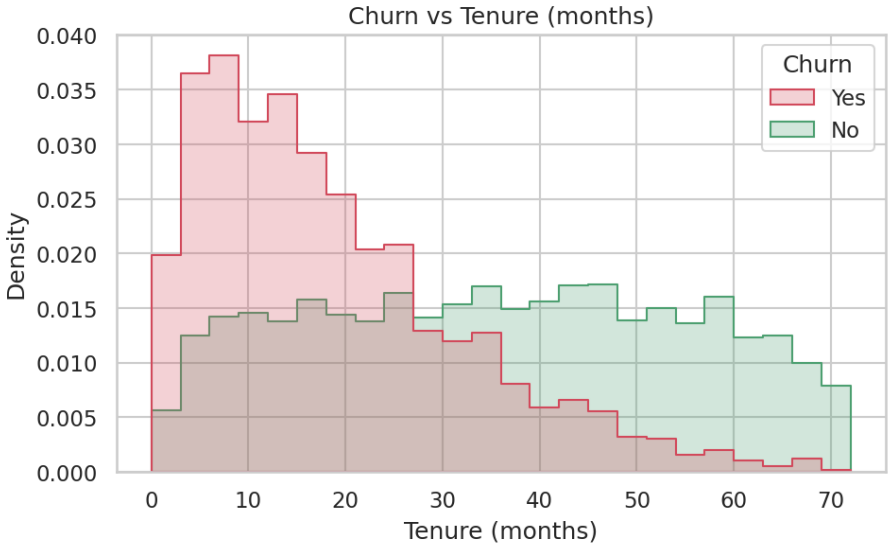


Figure 3 — Churn vs tenure: churners are concentrated among newer customers.

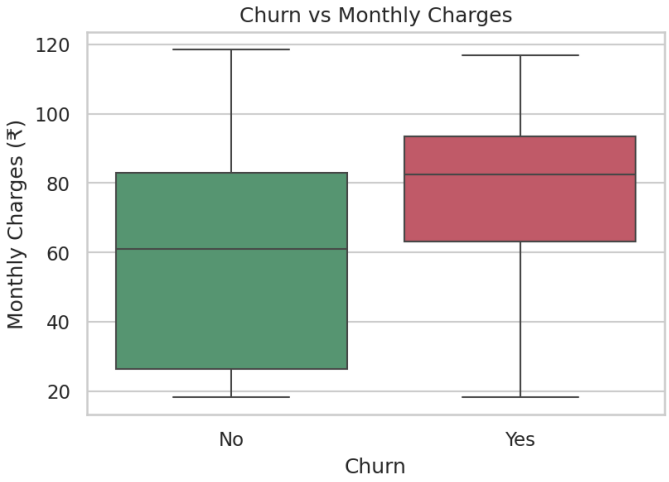


Figure 4 — Churn vs monthly charges.

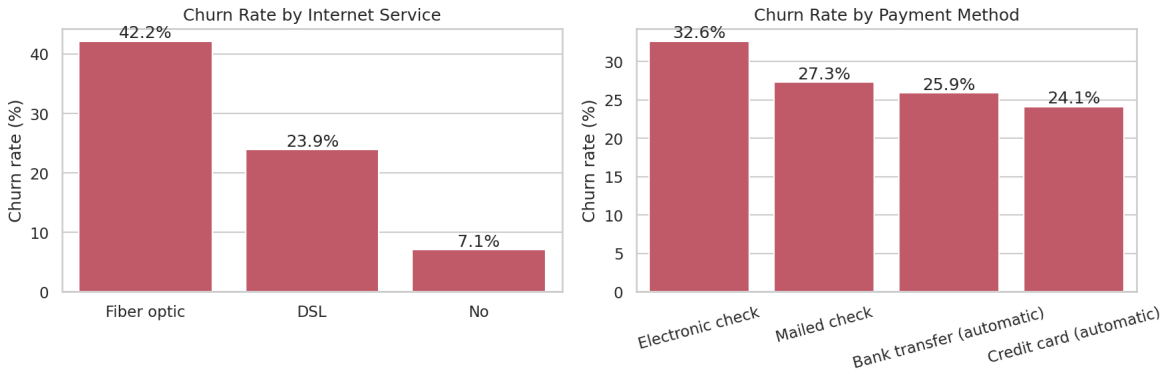


Figure 5 — Churn rate by internet service and payment method.

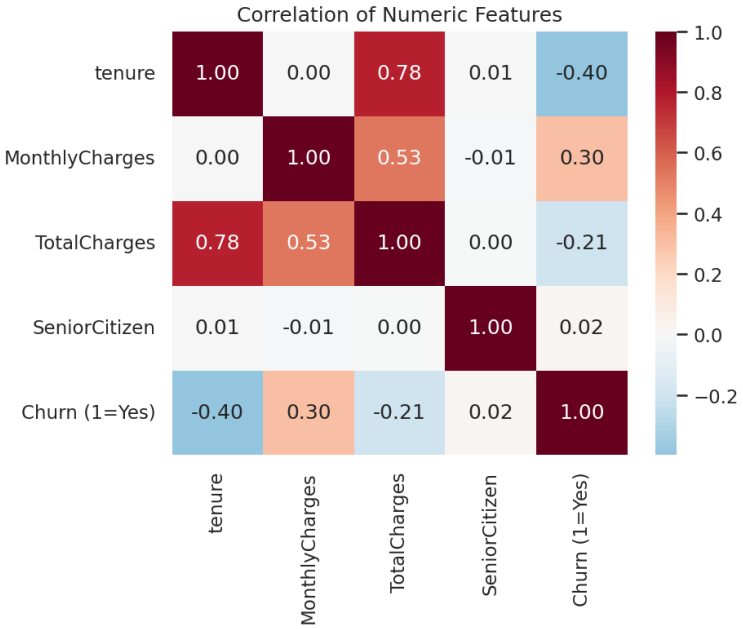


Figure 6 — Correlation of numeric features with churn.

9. Models Tested

Two baseline models were required; four further candidates were tested on Day 2 to improve the result:

- Logistic Regression — simple, fast, interpretable linear baseline.
- Random Forest (200 trees) — non-linear ensemble baseline. It reaches 100% accuracy on training data but less on test data, a clear sign of overfitting.
- Logistic Regression + engineered features + scaling.
- Logistic Regression + engineered features + class weighting (handles class imbalance).
- Gradient Boosting + engineered features (another algorithm).
- Random Forest + engineered features + class weighting, tuned with GridSearchCV (best parameters: max_depth=None, min_samples_leaf=8, n_estimators=200).

Candidates were ranked by 5-fold cross-validated F1 on the training data only; the test set was used purely to report performance.

10. Evaluation Results

Day 1 baseline comparison (test set)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.819	0.703	0.620	0.659	0.877
Random Forest	0.806	0.679	0.592	0.633	0.863

All experiments (selected model highlighted)

Experiment	CV F1	Acc.	Prec.	Recall	F1	AUC
1. Logistic Regression (baseline)	0.647	0.819	0.703	0.620	0.659	0.877
2. Random Forest (baseline)	0.603	0.806	0.679	0.592	0.633	0.863
3. Logistic Reg. + features + scaling	0.648	0.819	0.706	0.612	0.656	0.876
4. Logistic Reg. + features + class weight	0.674	0.790	0.596	0.793	0.680	0.877
5. Gradient Boosting + features	0.632	0.814	0.699	0.597	0.644	0.872
6. Random Forest + features + class weight (tuned)	0.671	0.791	0.600	0.781	0.678	0.869

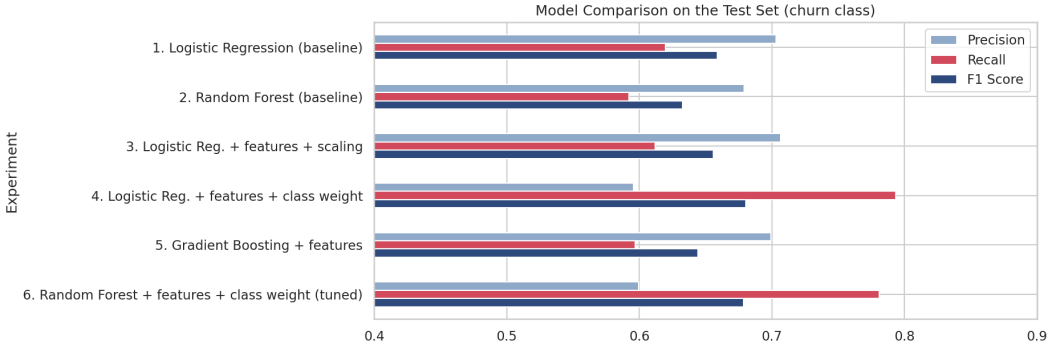


Figure 7 — Precision, recall and F1 of every experiment on the test set.

Which metric matters most?

Recall on the churn class, balanced against F1. A false negative (predicting 'stay' for a customer who leaves) silently loses the customer and all of their future revenue; a false positive only wastes a retention offer. Accuracy is not used to choose the model because only about 28% of customers churn, so a model that never predicts churn would already look accurate. F1 prevents 'buying' recall by flagging everyone.

Improvement documented (Challenge A)

	Model	F1	Recall	Precision
Before	Logistic Regression (baseline)	0.659	0.620	0.703
After	Logistic Reg. + features + class weight	0.680	0.793	0.595

What changed and why: class weighting was applied so the model treats a missed churner as more costly (the business requirement from Question 5), and engineered features and scaling were tested to summarise customer behaviour and fix the optimiser warning. Recall rose at the price of lower precision — more customers are flagged, some of whom would have stayed. That trade-off is a business decision.

11. Final Model

Selected model: Logistic Reg. + features + class weight. It was chosen by the highest cross-validated F1 on the training data and evaluated once on the 1,409 held-out customers.

Accuracy	Precision	Recall	F1 Score	ROC-AUC
0.790	0.595	0.793	0.680	0.877

	Predicted: Stay	Predicted: Churn
Actual: Stay	TN = 798	FP = 214
Actual: Churn	FN = 82	TP = 315

Of 397 customers who really churned, the model catches 315 (79.3%) and misses 82. It raises 214 false alarms among 1012 customers who stayed. TP = correctly flagged churner (can be saved); TN = correctly ignored loyal customer; FP = wasted retention offer; FN = silent loss (the most costly error).

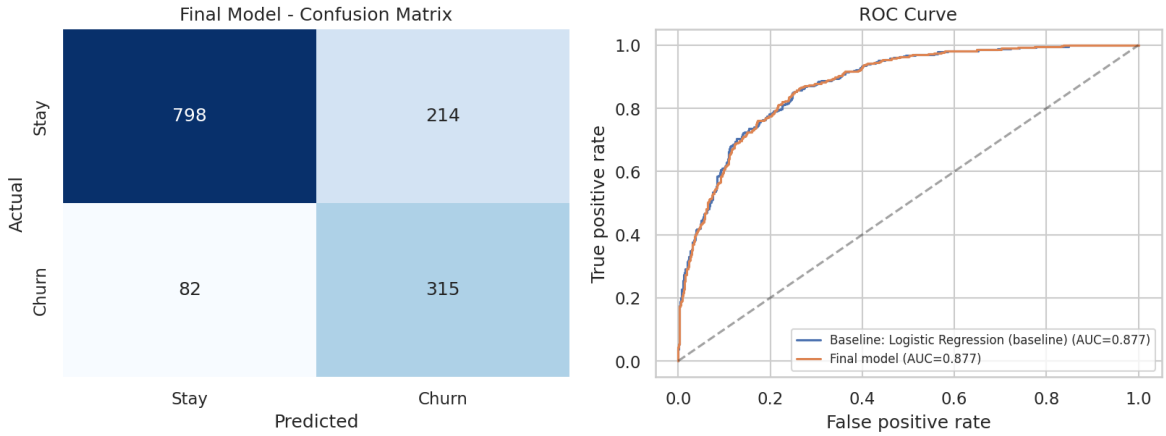


Figure 8 — Final model: confusion matrix and ROC curve (test set).

Feature importance

Permutation importance (drop in F1 when one original column is shuffled on the test set) shows what the model relies on:

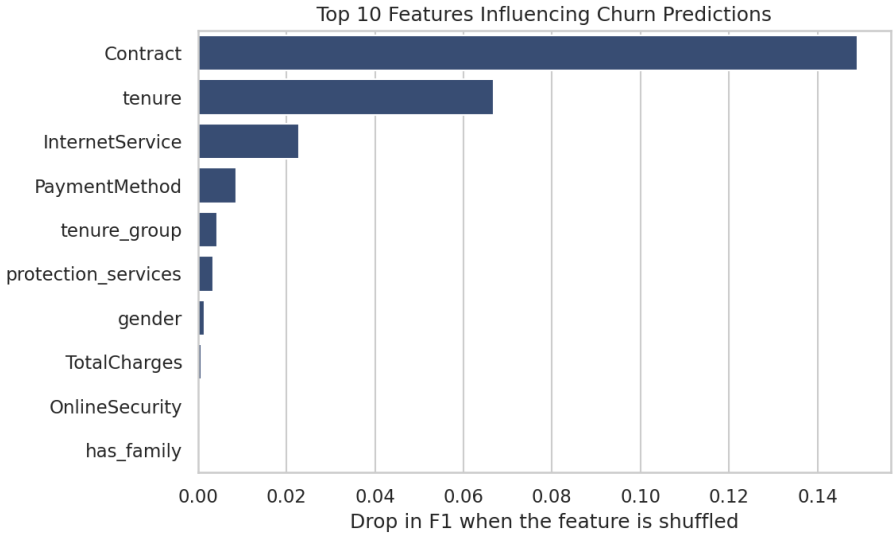


Figure 9 — Top 10 features influencing predictions.

Importance describes what the model uses, not why customers leave: it shows association, not causation.

12. Business Insights

- Contract type is the strongest business lever: month-to-month customers churn at 45.9% versus 3.5% on two-year contracts.

- Churn is concentrated among new customers: 51.4% churn in the first 12 months versus 4.4% after 4 years.
- The model relies mainly on Contract, tenure, InternetService; fibre-optic (42.2%) and electronic-check payers (32.6%) show above-average churn (overall 28.2%).

Risk levels (Challenge B)

Probabilities are converted to risk levels: HIGH ≥ 0.65, MEDIUM ≥ 0.40, otherwise LOW. The thresholds are a business choice justified by the churn rate actually observed inside each band on the test set (below). Because the final model is class-weighted, its probabilities are not calibrated frequencies and should be used for ranking.

Risk level	Customers	Share of customers	Actual churn rate	Churners captured
HIGH	414	29.4%	66.9%	69.8%
MEDIUM	205	14.6%	33.2%	17.1%
LOW	790	56.1%	6.6%	13.1%

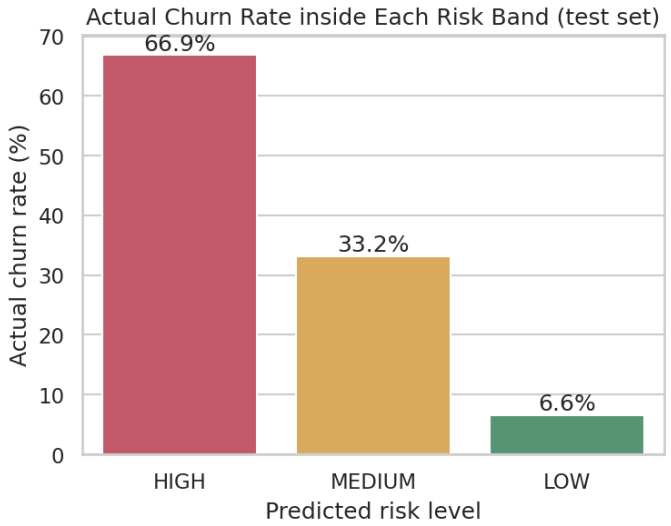


Figure 10 — Actual churn rate inside each predicted risk band.

Example predictions for new customers

Customer profile	Key inputs	Churn probability	Risk
New fibre customer on a monthly plan	Contract: Month-to-month; tenure: 5; MonthlyCharges: 79.5; InternetService: Fiber optic	93.3%	HIGH
Mid-tenure customer on a one-year plan	Contract: One year; tenure: 30; MonthlyCharges: 65.0; InternetService: DSL	19.2%	LOW
Loyal customer on a two-year plan	Contract: Two year; tenure: 60; MonthlyCharges: 55.0; InternetService: DSL	1.8%	LOW

Recommendation

Route customers scored HIGH risk (probability >= 0.65) to a retention team; offer discounted 1-2 year contracts to month-to-month customers, build a first-12-months onboarding programme, investigate fibre-optic service quality, and encourage automatic payment methods and security/support add-ons.

- HIGH risk → personal outreach or targeted offer.
- MEDIUM risk → automated nudges and service check-ins.
- LOW risk → routine engagement only.

13. Limitations

- The dataset used here is a schema-identical replica of the IBM Telco data - re-run with the official CSV to obtain official metrics.
- Single historical snapshot: no time dimension, so seasonality and trends are not captured.
- The model shows association, not causation - a discount is not guaranteed to change behaviour.
- Probabilities from a class-weighted model are not calibrated; use them for ranking, not as exact percentages.
- Retention cost / customer lifetime value were not in the data, so the ideal risk threshold is a business assumption.
- A prediction is not certainty: a share of flagged customers would not have left.

14. Future Improvements

- Add a cost-based decision threshold using real retention-offer cost and customer lifetime value.
- Test gradient-boosting libraries (XGBoost / LightGBM) and probability calibration.
- Explain individual predictions with SHAP values.
- Use time-stamped data and survival analysis to predict when a customer is likely to leave.
- Run an A/B test of retention offers to measure real (causal) uplift.
- Deploy the Streamlit app, monitor data drift and retrain regularly.

15. Project Evidence Checklist

Required evidence	Where to find it
Dataset preview	Section 4 of this report; notebook Step 4 (df.head())
EDA visualisations	Figures 1-6; images/01 ... 06; notebook Step 6
Model evaluation	Evaluation Results tables, Figure 7; notebook Steps 10-12 and Challenge A
Confusion matrix	Figure 8; images/07 and images/09; notebook Step 13 and Challenge A
Final prediction	Business Insights (example predictions); notebook Steps 14 and Challenge B
Final result	Final Model section; notebook Challenge C ('CHURNGUARD AI — FINAL RESULTS')